

The Economics of a 3D Print Farm

From machine-hour costing to a Redwood City business plan — the mathematical case, its formulas, and what they resolve to

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Provenance: Synthesizes a five-specialist accounting panel with adversarial credit-officer review, a deep-research pass (105 agents, 3-vote adversarial claim verification), and three targeted fact-checking sweeps of Redwood City / California / unit-economics inputs. Claims that failed verification are listed in §9.3 and are not used anywhere in this report.

Status: Planning analysis — not tax, legal, or investment advice.

1 Executive summary — five equations, five conclusions

This report converges the full analysis into its quantitative skeleton: each business conclusion is stated with the formula that produces it, the reason that formula is the right tool, and the verified 2026 inputs that make it resolve the way it does.

The equation	What it computes	What it resolves to
$MHR(H) = F/H + v$ (§2.1)	Fully absorbed machine-hour rate	$\approx \$0.25\text{--}0.34/\text{hr}$ — the machine is nearly free; hardware is <i>not</i> the business
$CM/h = (p - c_v)/t_m$ (§3.2)	Contribution margin per constrained machine-hour	Product ranking <i>inverts</i> : the best product per unit is the worst per hour
$U^*(c) = M/(c \cdot H)$ (§4.3)	Break-even utilization for a required cash burden M	Loan alone: $\sim 6\%$. Loan + owner's living wage: 34% — rising to $\sim 68\%$ at realistic day-one margins. The owner's wage, not the loan, is the hurdle
$NPV = -K + \sum CF_t(1+d)^{-t}$, with CF_t net of $w_f h_f$ (§5)	Investment value <i>after imputed founder wage</i>	Positive only with differentiated (service/software) margins; hardware-utilization models fail
$EP = OP - w_f h_f - \kappa K$ (§5.3)	Economic profit	Year 1 is negative by design; the purchase is information + IP + optionality at a capped $\sim \$4\text{--}6k$ worst case

The one-paragraph verdict. Verified industry data (Wohlers 2026: services are 48% of the \$24.2B AM market and growing 15.5%/yr versus 3.6% for machine sales) plus verified operator economics (\$75–400/printer/month gross, base case) prove that a print business survives on **demand relationships, speed, and software** — not on machine ownership. The plan that follows from the math: a home-based, software-leveraged, same-day B2B prototyping cell in Redwood City, \$10k founder capital, milestone-gated debt (SBA microloan, 8–13%), equity only for software traction, and a founder-wage-honest scorecard that decides scale-up or shutdown at months 4, 9, and 15.

2 The cost engine

2.1 The machine-hour rate and the absorption denominator

Every job must absorb a share of the printer's fixed annual pool F (economic depreciation, insurance, software, space share) plus the variable running cost v (electricity + wear-driven maintenance) for each hour it occupies the machine:

$$\text{MHR}(H) = \frac{F}{H} + v \quad (1)$$

Why this form: fixed costs are incurred per *year* while jobs consume *hours*, so the only economically meaningful bridge is division by the annual hour base H . Because H sits in a denominator, the rate is a **hyperbola** in utilization — cheap at high volume, explosive at low volume (Figure 1).

How it resolves: with verified P1S-class inputs ($F \approx \$625/\text{yr}$, $v \approx \$0.095/\text{hr}$), $\text{MHR} = \$0.41$ at 2,000 hr/yr, **\$0.25 at 4,000**, \$0.20 at 6,000. Two conclusions follow directly: (i) the machine contributes only ~17% of a typical job's cost — electricity ~1% — so hardware and power are the wrong optimization targets; (ii) H must be set at **practical capacity**, never at current actual volume. Setting H to a low actual volume inflates the quoted rate, which loses bids, which lowers volume further — the classic absorption **death spiral**, visible as the left region of the curve.

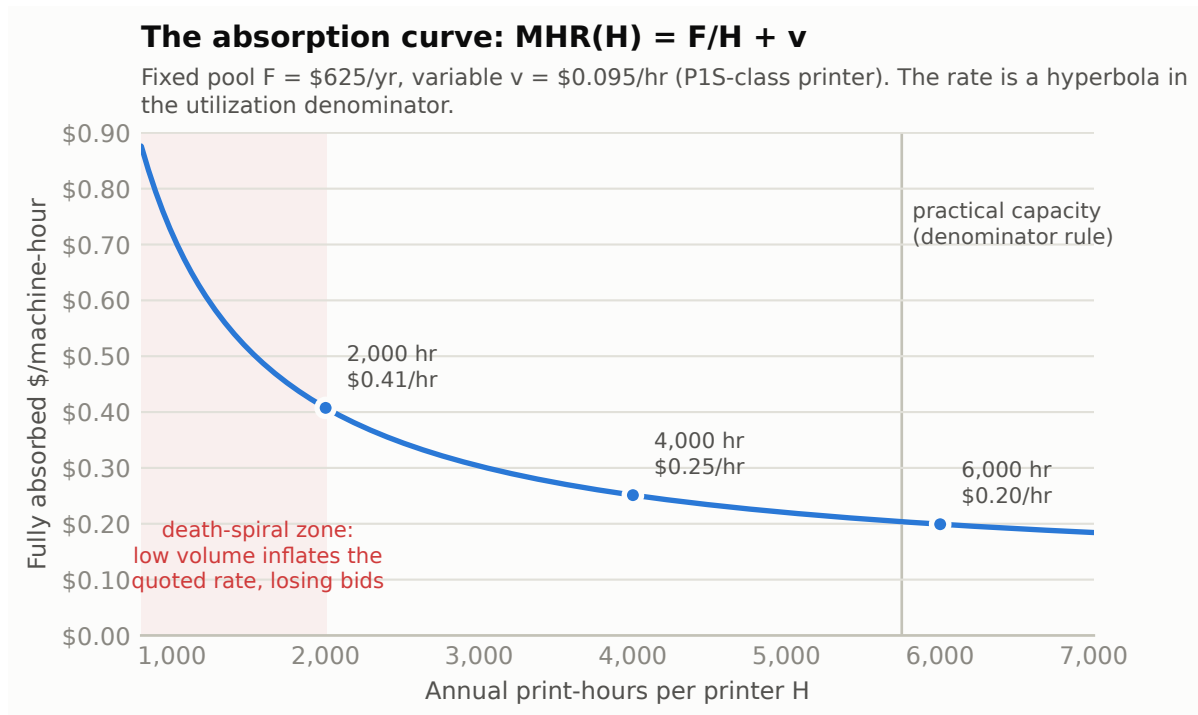


Figure 1. Equation (1) with $F = \$625/\text{yr}$, $v = \$0.095/\text{hr}$. The marked points are the quoting rates at 2,000 / 4,000 / 6,000 annual print-hours. Quoting from the left side of the curve is self-reinforcing decline; the dashed marker shows the practical-capacity denominator rule.

2.2 Job standard cost with a spoilage load

A job consuming m grams of material at price p_m (with purge/support waste fraction ω), t machine-hours, and ℓ minutes of touch labor at loaded rate r_L , with a normal scrap rate s , has standard cost:

$$C_{\text{job}} = [m(1 + \omega)p_m + t \cdot \text{MHR}] \cdot (1 + \frac{s}{1-s}) + \ell \cdot r_L \quad (2)$$

Why the multiplicative load $s/(1-s)$: if a fraction s of attempts fail, producing one *good* unit requires on average $1/(1-s)$ attempts; the extra $(1/(1-s) - 1) = s/(1-s)$ attempts consume material and machine time but not (all) packing labor. At the professional-farm standard $s = 4\%$, the load is $+4.2\%$. Failures beyond the standard are **abnormal spoilage** — a separate expense line, never buried in product cost, because burying them silently corrupts every future quote.

How it resolves (140 g PETG part, 5.5 hr, honest 22 min touch labor at \$30/hr loaded): $C \approx$ **\$16.89, of which labor is ~65%**. The margin battle is labor minutes; each 5 minutes automated out of a job adds $\approx$ \$2.50 of margin at zero marginal cost. This is the arithmetic reason the software layer (\$7) *is* the business.

2.3 Failure risk compounds with print duration

Print failures arrive approximately as a constant per-hour hazard λ , so survival over a t -hour job is exponential:

$$P_{\text{fail}}(t) = 1 - e^{-\lambda t} \quad \Rightarrow \quad \text{CM}_{\text{risk}} \approx \text{CM} \cdot e^{-\lambda t} \quad (3)$$

Why exponential: a memoryless per-hour failure chance (clogs, spaghetti, power blips) is the natural null model, and it matches the observed pattern that long prints fail disproportionately.

How it resolves: at the industry-typical $\lambda \approx 0.5\%/hr$, an 8-hour job fails $\sim 4\%$ of the time but a 50-hour job fails **$\sim 22\%$** (Figure 2) — and a failure at hour 45 destroys 45 machine-hours of capacity, not just \$5 of plastic. Consequences: segment long prints into shorter plates, first-layer-check anything > 8 hr, and risk-discount long-print products in the ranking of §3.2 (this is what demotes the turbine-class product).

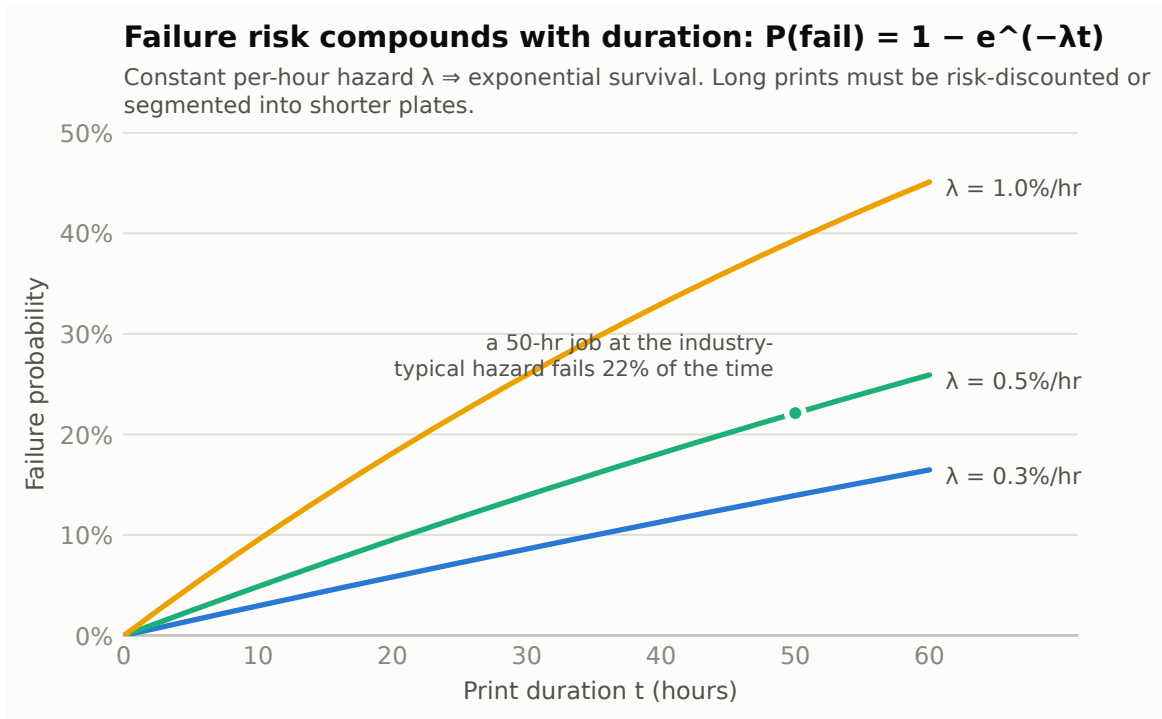


Figure 2. Equation (3) at three hazard rates. The verified professional band is 2-5% *per attempt* on normal jobs; duration converts a small hourly hazard into a large per-job risk for long prints.

3 Capacity and throughput

3.1 TEEP: how calendar hours become good hours

$$\text{TEEP} = L \times A \times P \times Q \quad G = 730 \cdot \text{TEEP} \quad \text{good hr/printer/month} \quad (4)$$

Why TEEP and not OEE: loan interest, insurance, and obsolescence accrue on all 8,760 calendar hours whether or not hours were "scheduled," so the honest denominator is the calendar (TEEP), not the schedule (OEE). The four factors decompose loss into its owners: **L**oading (does the queue hold a job? — the demand factor), **A**vailability (changeovers, jams), **P**erformance (actual vs. standard speed), **Q**uality (first-pass yield).

How it resolves: a well-run base case $L = 71\%$, $A = 90\%$, $P = 90\%$, $Q = 95\%$ gives $\text{TEEP} \approx 55\% \Rightarrow G \approx 400 \text{ good hours/printer/month}$ — squarely inside the verified 50–70% industry band, while manual-only operation runs 24–30%. The decomposition shows **Loading is almost always the binding loss:** the constraint is demand, not machinery — the single most important fact in this report. Product variety earns its keep here (filling the queue) and only here; it is managed at the queue level while daily plates stay batched.

3.2 Ranking products: contribution margin per machine-hour

$$\frac{\text{CM}}{h} = \frac{p - c_v}{t_m} \quad (5)$$

Why divide by machine time: theory of constraints — when the queue is full, the machine-hour is the scarce resource, and the value of a product is the margin it earns *per unit of the constraint it consumes*. Margin per unit ignores the denominator and therefore systematically misallocates the constraint.

How it resolves (Figure 3): the three archetype products invert. The turbine-class assembly leads per unit (\$125) and finishes last per hour (\$2.50, ≈\$2.00 after the 22% failure exposure from Eq. 3); the humble fast-turn frame-class product wins at \$4.30/hr. Operating rule: rank the whole catalog by risk-adjusted CM/h, schedule strictly by that ranking, hold an external floor (≈\$3/hr in the Redwood City plan), and let long low-CM/h prints run only in otherwise-idle overnight slots priced against the ≈\$0.12 variable floor.

The throughput inversion: the best product per unit is the worst per constrained machine-hour

Rank by CM/h = $(p - c_v) \div t_{\text{machine}}$ when the machine is the constraint (theory of constraints).

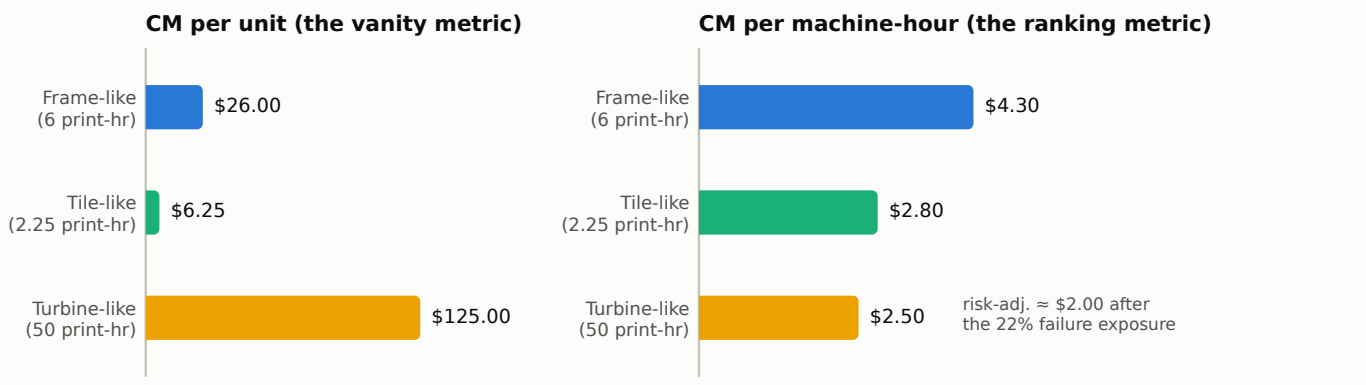


Figure 3. The same three products under the two metrics. Left: margin per unit (the vanity metric). Right: Equation (5). The constrained-resource ranking reverses the intuitive one.

3.3 Operator leverage: step-fixed labor

$$\text{labor cost per good hour} = \frac{W}{N \cdot G} \quad (6)$$

Why: an operator is a step-fixed cost W spread over the good hours of the N printers they can supervise; automation raises N . **How it resolves** (Figure 4): at $W = \$52k$ and $G = 4,800$ good hr/yr, labor per good hour falls $\$10.83 \rightarrow \$2.17 \rightarrow \$1.08 \rightarrow \0.54 across $N = 1, 5, 10, 20$. Below ~ 5 printers the economics work only because the owner's labor is unpriced; the efficient scale unit is the **10-15 printer block that saturates one operator** — which is why the plan's debt tranche (§7) is sized to that block and why financing one or two premium machines is the worst possible structure.

Step-fixed labor dilution: $\text{labor/hr} = W \div (N \cdot G)$

$W = \$52\text{k}$ loaded operator cost; $G = 4,800$ good hr/yr per printer at 55% TEEP. Labor, not the machine, dominates unit cost at small N .

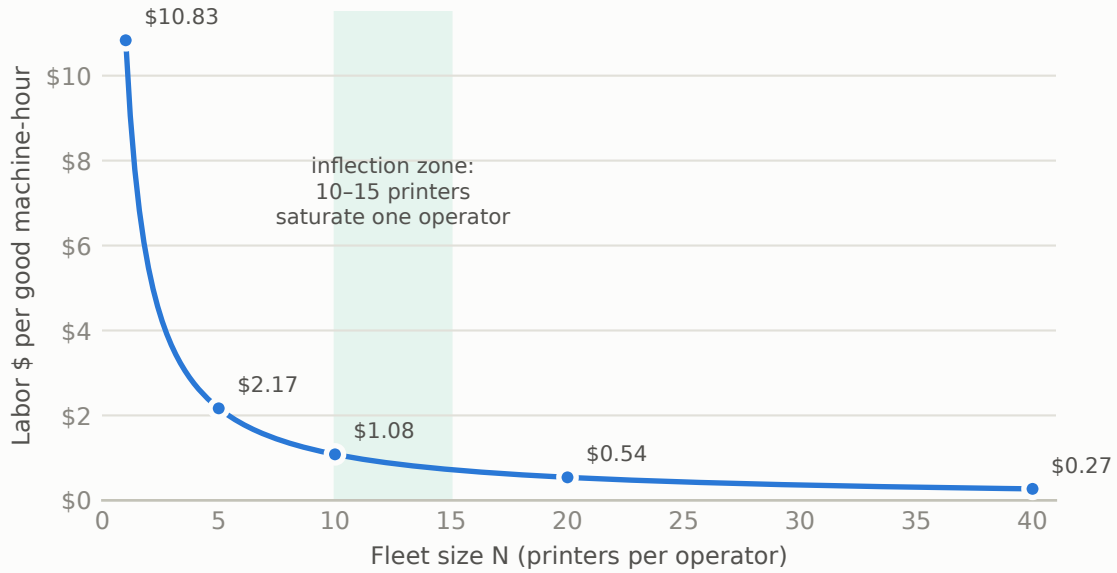


Figure 4. Equation (6). Labor — the largest controllable cost in §2.2 — is diluted only by fleet size per operator, which is a software/automation variable, not a hardware one.

4 The loan mathematics

4.1 Amortization

$$A = P \cdot \frac{r}{1 - (1 + r)^{-n}} \tag{7}$$

Why: the level payment A that repays principal P over n months at monthly rate r is the annuity solution to the loan's present-value identity. **How it resolves:** a \$20,000 SBA-microloan-class note at 11%/72 months gives **$A = \$380.66/\text{month}$** (total interest $\approx \$7,408$); the verified program band is 8–13%, up to \$50k, average $\sim \$13\text{--}16\text{k}$ — exactly this deal's size. Discipline: term $\leq$ the asset's ~ 36 -month competitive life. Ten-year money on three-year printers means paying in years 8–10 for machines that died in year 4.

4.2 Debt service coverage — after the owner's wage

$$\text{DSCR} = \frac{\text{CFADS}}{12 \cdot A} \geq 1.25 \quad (\text{bank}) \quad \geq 1.5 \quad (\text{internal, after market owner wage}) \tag{8}$$

Why the wage adjustment: cash flow available for debt service that exists only because the owner works unpaid is not repayment capacity — it is the owner personally guaranteeing the loan with foregone wages. Lenders underwrite small notes on *global* cash flow (household income, credit score) for exactly this reason; the printers themselves collateralize perhaps 10–15% of the note (40–60% resale haircut within two years, thin secondary market).

4.3 Break-even utilization — the equation that decides everything

Let M be the monthly cash burden to be covered (debt service, fixed overhead, and — honestly — the owner's draw), c the blended CM per machine-hour (Eq. 5), and H the fleet's calendar machine-hours per month. The utilization that covers M is:

$$U^*(c) = \frac{M}{c \cdot H} \tag{9}$$

Why this is the master sensitivity: every risk in the business — price erosion, failure rate, product mix, demand — enters through either M or c , and U^* makes their joint effect visible on one axis against the physically attainable band. **How it resolves** (10-printer farm, $H = 7,300$ hr/mo; Figure 5):

Burden covered	M (\$/mo)	U* at c = \$2.00	U* at c = \$1.00 (day-one commodity)
Debt service + overhead only	881	6%	12%
+ DSCR 1.25 + \$4,000/mo owner draw	4,976	34%	68%

The loan line is trivially low — which is precisely why "the printer pays for itself" is a true statement and a worthless one. The owner-wage line at realistic day-one margins sits **at or above the 50-70% ceiling of well-run farms**. Conclusion, direct from Eq. (9): the loan is never broken by interest; it is broken by *mix* (c) and by pretending the owner works for free (M). A commodity-only fleet caps out at $c \cdot H \approx \$2,900/\text{month}$ at an impossible 100% utilization — structurally unable to support an owner at any interest rate, including 0%.

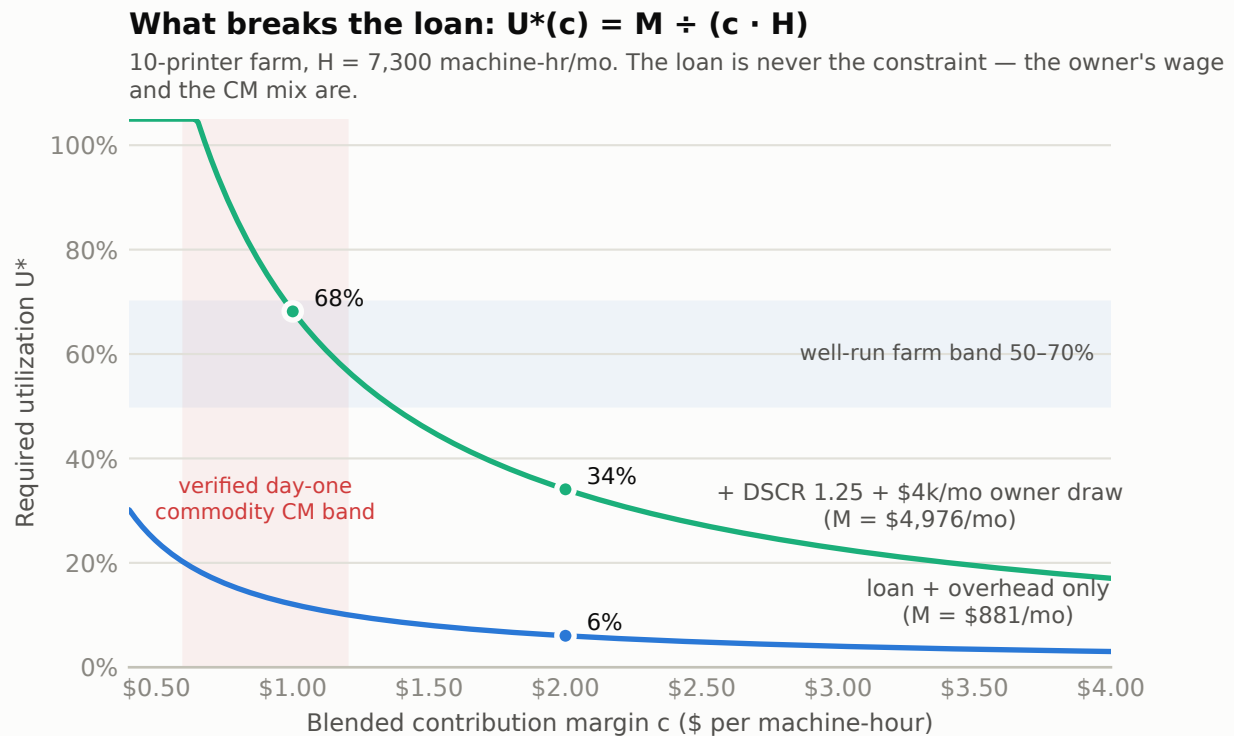


Figure 5. Equation (9) for both burden levels. Shaded horizontal band: attainable utilization for well-run farms (50–70%). Shaded vertical band: the verified day-one commodity CM range (\$0.60–1.20/hr). The green curve crossing above the band inside that range *is* the case against borrowing for undifferentiated capacity.

Lending standard that follows: borrow only against demand evidence that moves c out of the commodity band — a signed B2B PO covering ≥ 12 months of debt service, 3+ months of $\geq \$2\text{--}3\text{k}/\text{month}$ revenue at $\geq 40\%$ CM, or deposits $\geq 25\%$ of principal — and size the note so $\text{DSCR} \geq 1.5$ holds *after* a $\$25\text{--}30/\text{hr}$ owner wage, stress-tested at $c = \$1.00\text{--}1.25$.

5 Investment value and owner economics

5.1 NPV, corrected for unpaid labor

$$\text{NPV} = -K + \sum_{t=1}^T \frac{\text{CF}_t}{(1+d)^t}, \quad \text{CF}_t = \text{OP}_t - w_f h_f \quad (10)$$

Why the correction matters: the adversarial review of this analysis found the single largest error in the original panel was an NPV of +\$65k computed on cash flows that priced the founder's 200–300 hours/month at zero — the wage bill alone exceeded the three-year contribution. Once CF_t is net of the imputed wage $w_f h_f$, the base-case hardware-utilization NPV turns **decisively negative**, while service/software-weighted cases (higher c , lower h_f through automation) remain positive. An NPV that only works at $w_f = 0$ is not a business; it is a job bought with borrowed money.

5.2 Cost of capital anchors

$$r_{\text{after-tax}} = r(1 - \tau) \quad (11)$$

Interest is fully deductible at this scale (§163(j) small-business exemption), so an 11% note at a ~30% combined marginal rate costs $\approx 7.6\%$ after tax — the hurdle each marginal machine's contribution must clear. Equity has no coupon but is the most expensive money available; spending it on depreciating commodity hardware (–40–60% in two years) is arithmetic value destruction, which is why the capital plan reserves equity exclusively for software traction.

5.3 Economic profit and the owner's wage test

$$\text{EP} = \text{OP} - w_f h_f - \kappa K \quad w_{\text{implied}} = \frac{\text{OP} - \kappa K / 12}{h_f} \quad (12)$$

Why two forms: EP answers "is this creating wealth?"; the implied wage answers "what is this paying me per hour, really?" — the honest number an owner compares to their market alternative (here \$70–100+/hr for a Bay Area data scientist). **How it resolves** for the Redwood City plan: base-case month 12 gives $\text{OP} \approx \$1,150/\text{mo}$ on ~40 founder-hours $\Rightarrow$ implied wage $\approx \$29/\text{hr}$; with $\kappa = 10\%$ on $K = \$10\text{k}$, year-one EP is negative. The plan states this openly and prices what year one actually buys: verified market information, reusable software IP, a B2B relationship set, and optionality — at a capped worst case of ~\$4–6k after equipment resale. The gate structure (§7.4) converts EP from a verdict into a decision rule: Gate D fires when the implied wage sustains $\geq \$70/\text{hr}$.

6 Tax mathematics (federal / California, 2026 — verified)

6.1 Two depreciation clocks and the timing cost

Federal law (OBBA, permanent) allows 100% first-year expensing of the fleet (bonus depreciation, or de minimis expensing for sub-\$2,500 items). California conforms to neither bonus depreciation nor the federal §179 limit — CA §179 is capped at \$25,000 and usable only against business income, so a loss-year startup falls back to MACRS. The present-value cost of California's slower clock is:

$$\Delta PV = \sum_{t=1}^8 \frac{(d_t^F - d_t^{CA}) \cdot \tau_{CA}}{(1 + d)^t} \quad (13)$$

How it resolves (Figure 6): on \$15,000 of printers in a loss year, the year-1 California deduction is \$2,144 (7-yr MACRS half-year, 14.29%) versus \$15,000 federal — a \$12,856 timing gap reversing over eight years, worth roughly \$1,100–1,200 of deferred CA tax benefit at a 9.3% marginal rate. Planning consequences: keep two depreciation schedules from day one; in a *profitable* year, elect CA §179 (the fleet is under the \$25k cap) and the gap disappears; and never let the federal write-off masquerade as free equipment — the management P&L (\$2) still charges ~\$0.25/machine-hour of real capital consumption and funds a replacement reserve.

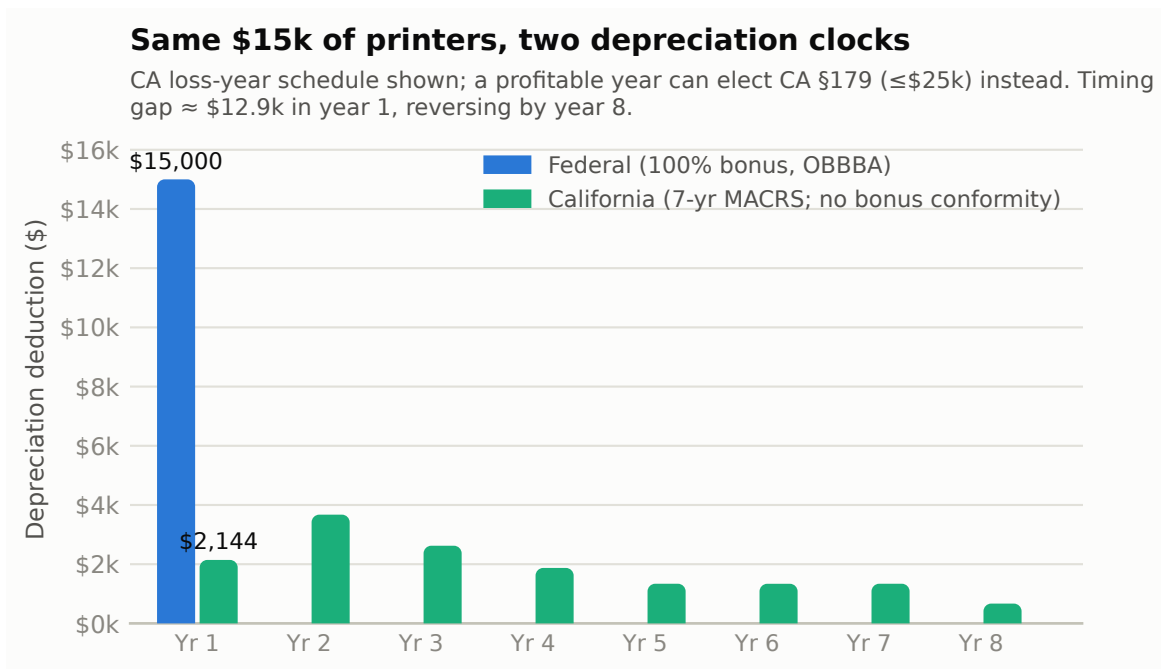


Figure 6. The same \$15k asset under the two clocks (CA loss-year case). The bars are equal in total; only timing differs — Equation (13) prices that timing.

6.2 The transaction-tax layer that shapes pricing

- **All fabrication revenue is taxable** at Redwood City's 9.875% (CDTFA Pub 108) — itemized or bundled, even on customer-supplied material; design-and-production contracts are taxable on the entire price including separately stated design (Reg. 1501.1). Sales tax is collected in trust and never counted as revenue.
- **Inputs:** filament tax-free with a resale certificate; printers qualify for the manufacturing partial exemption (RTC 6377.1, through 6/30/2030): the effective rate on qualifying equipment drops 9.875% → 5.9375%.
- **Structure:** \$800/yr CA LLC franchise tax from year one (waiver expired); no CA QBI deduction (unlike the permanent federal 20%); S-corp election is net-negative below ~\$90–110k sustained profit once payroll costs and QBI erosion are counted.

7 Application: Peninsula Additive (Redwood City, July 2026)

7.1 The strategy the math selects

Feeding the verified local inputs into §§2-6 selects one strategy dominantly: a **home-based, software-leveraged, same-day B2B prototyping cell**. The reasoning chain, formula by formula:

- 1. **Eq. (1)** says hardware is ~17% of job cost $\Rightarrow$ owning printers is not the edge; Redwood City's ~\$0.40/kWh electricity (2.5-3 $\times$ national average) raises v by only ~\$0.03/hr — immaterial.
- 2. **Eq. (2)** says labor is ~65% $\Rightarrow$ the founder's automation skill attacks the dominant cost directly.
- 3. **Eq. (5)** + verified prices (JLC3DP from \$0.30/part; bureaus \$8-40 per 100g part; free library PLA locally) $\Rightarrow$ commodity work sits in the fatal CM band of Figure 5; the only verified premium is **speed + bundled design labor (\$40-125/hr)** — deliverable locally, impossible for ship-in networks.
- 4. **Eq. (9)** $\Rightarrow$ no debt at day one: at \$10k total capitalization the burden M stays small, and the wage-honest utilization hurdle is unreachable with commodity c anyway.
- 5. Local constraints close the choice: industrial space $\approx$ \$1,650-2,000/mo + NNN (kills leased space at this scale); home-occupation rules cap the fleet at a quiet spare-room cell — the correct phase-one size regardless.

7.2 Deploying the \$10,000

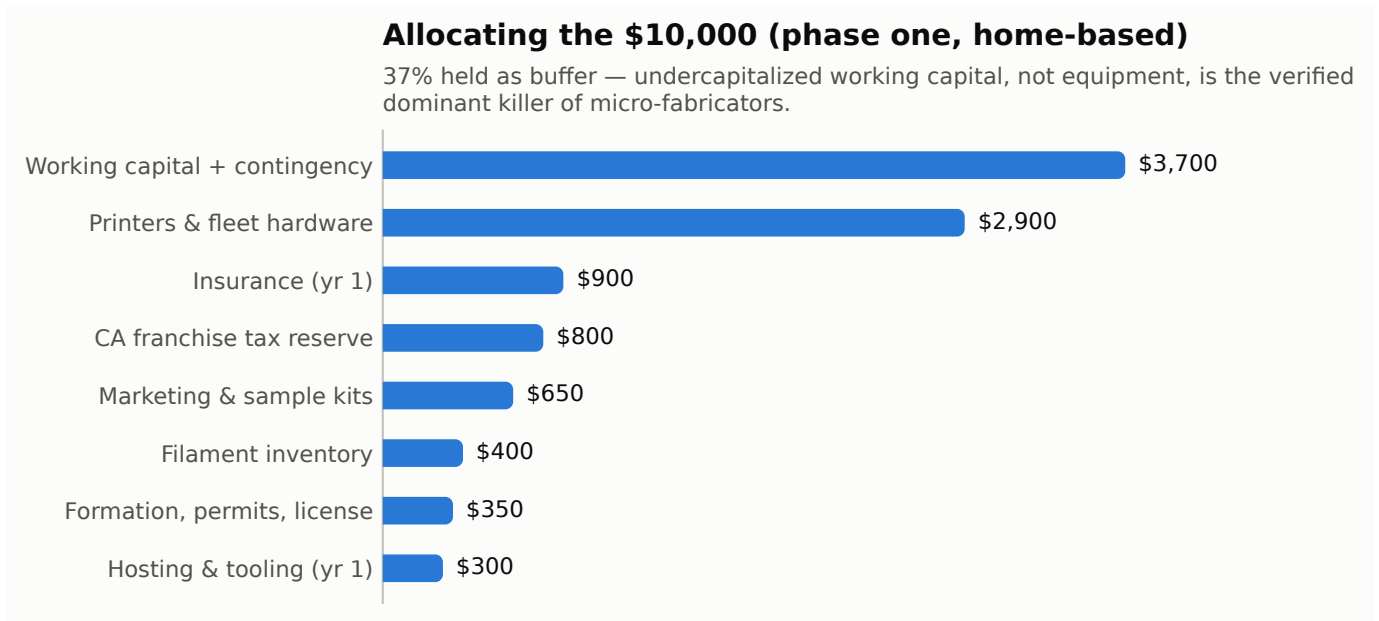


Figure 7. Phase-one allocation. 37% is working-capital buffer — verified operator history shows cash starvation, not equipment shortage, as the dominant micro-fabricator killer. Absent lines: rent (home-based), software licenses (founder-built), employees (Redwood City wage floor \$18.65/hr defers hiring to ~\$8k/mo revenue).

7.3 Year-1 trajectory and scenarios

Revenue anchors are deliberately mid-band: printing reaches \$325/printer/month by month 12 — the *middle* of the verified \$75–400 operator range — plus 11 design-labor hours/month at \$75/hr. COGS is modeled at 32% of printing revenue with \$290/month of fixed costs (insurance, franchise-tax accrual, hosting, license).

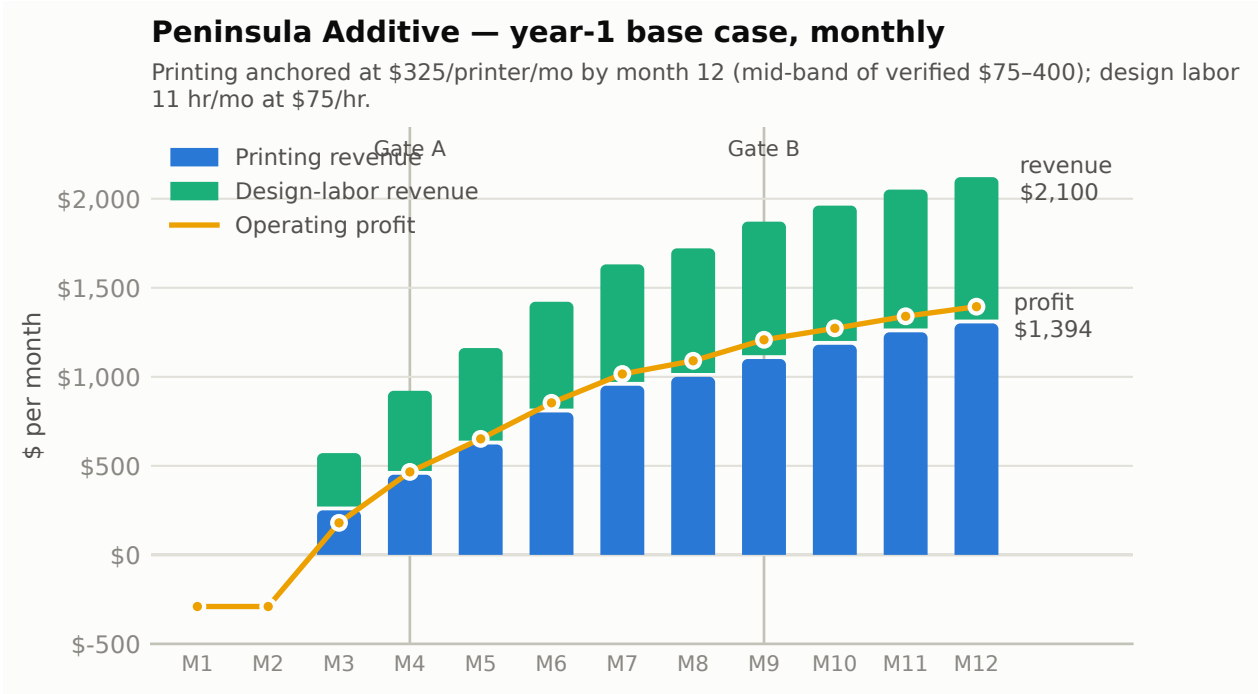


Figure 8. Base-case months 1–12. Gate A (month 4) tests demand signal: ≥ 10 paying B2B jobs, ≥ 3 repeat accounts, $CM/h \geq \$3$. Gate B (month 9–12) tests scale-readiness: 3 consecutive months $\geq \$2,000$ revenue at $\geq 40\%$ CM.

Year-1 scenarios (before founder compensation)

Month-12 per-printer gross: \$120 / \$325 / \$700 — bottom, middle, and top of the verified operator band.

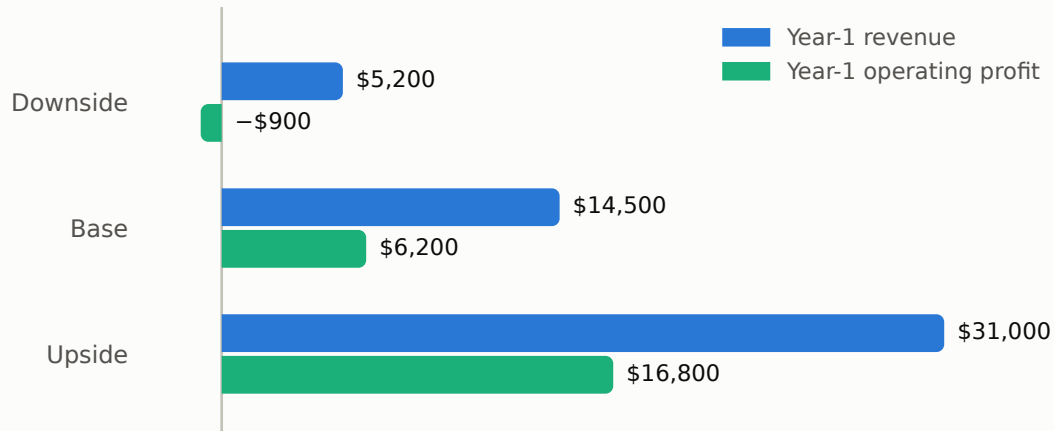


Figure 9. The three scenarios differ almost entirely in the demand variables (per-printer gross and design hours) — the capital cost is identical, which is Figure 5's lesson restated at plan scale: variance is ~98% demand-driven, ~2% CapEx-driven.

7.4 Milestone-gated capital plan

Gate	When	Quantitative test	Pass →	Fail →
A — signal	Month 4	≥10 paid B2B jobs; ≥3 repeats; CM/h ≥ \$3	Add printer 4 from cash flow	Pivot to design/software-only, or wind down (~\$5–6k recovered)
B — scale	Months 9–12	3 consecutive months ≥\$2,000 rev at ≥40% CM; backlog ≥5 days; yield ≥95%	SBA microloan \$15–25k (Eq. 7: ≈\$330–545/mo); 6+ printers; space decision	Hold at 4 printers — a stable side asset is a legitimate steady state
C — software	Opportunistic	≥3 external operators using the tooling, asking to pay	Productize; consider SAFE \$50–150k	Keep internal as a margin weapon
D — commitment	Any time	Implied wage (Eq. 12) ≥ \$70/hr for 2 consecutive quarters	Full-time decision	Keep the portfolio posture

The owner's covenant: demand evidence precedes every tranche; debt (7.6% after-tax, Eq. 11) before equity; equity never buys hardware that loses half its value in two years; DSCR ≥ 1.5 after a market owner wage at every step (Eq. 8); household runway independent of the business throughout year one.

8 Conclusions — from formula to decision

Formula	What it proved	Decision it forces
(1) $MHR = F/H + v$	Machine $\approx$ \$0.25–0.34/hr; power $\approx$ 1–2% of job cost	Stop optimizing hardware and electricity; quote at practical capacity; never enter the absorption death spiral
(2) job cost with $s/(1-s)$	Labor $\approx$ 65% of job cost	Automate labor minutes (the founder's edge); separate abnormal spoilage; 4% scrap standard
(3) $P = 1 - e^{-\lambda t}$	50-hr prints fail $\sim$ 22% at typical hazard	Segment long jobs; risk-discount long-print products; first-layer checks >8 hr
(4) $TEEP = L \cdot A \cdot P \cdot Q$	Loading (demand) is the dominant loss factor	Sell before building; variety at queue level; lights-out unlock first
(5) $CM/h = (p - c_v)/t_m$	Product ranking inverts under the constraint	Rank catalog by risk-adjusted CM/h; \$3/hr external floor; overnight slots for low-CM/h filler
(6) $W/(N \cdot G)$	Labor dilutes only with printers per operator	Scale in 10–15-printer operator-saturating blocks — never one premium machine on debt
(7)–(9) A, DSCR, $U^*(c)$	Loan servicing needs $\sim$ 6% utilization; owner wage needs 34–68%	Borrow against contracts/run-rate, not capacity; term $\leq$ asset life; underwrite after owner wage at stressed c
(10) NPV net of $w_f h_f$	Unpaid-labor NPVs are illusions	Reject any plan positive only at a \$0 founder wage
(12) EP, implied wage	Year-1 EP negative; implied wage $\approx$ \$29/hr	Treat year 1 as a capped-cost option; Gate D converts the wage test into the full-time decision
(13) ΔPV of CA timing	CA defers $\sim$ \$12.9k of year-1 deductions in a loss year	Two depreciation schedules; CA \$179 in profitable years; management P&L with replacement reserve

The unifying result: every formula, fed the verified 2026 inputs, points away from "buy machines, find demand later" and toward "prove demand with software-leveraged speed, then let cash flow — and only then cheap, short, demand-backed debt — buy machines." That is the mathematically justified formulation of the original thesis.

9 Appendix — inputs, verification, and refuted claims

9.1 Key verified inputs used in this report

Input	Value	Source class
Global AM market 2025 / services share & growth	\$24.2B; 48% / +15.5%	Wohlers Report 2026 (primary)
P1S-class printer, street / all-in	~\$539 / ~\$800	Vendor + price history
Fixed pool F / variable v (per printer)	\$625/yr / \$0.095–0.14/hr	Component build-up, \$2.1
Operator revenue per printer (base / top-decile)	\$75–400 / ~\$1,200 mo	Operator-reported, multi-source
Bureau price, ~100g FDM part / China floor	\$8–40 / from \$0.30	Marketplace-observed
Design/engineering labor band	\$40–125/hr	Marketplace (Upwork/Fiverr/salary data)
Well-run farm utilization band	50–70% (manual 24–30%)	Industry benchmarks
SBA microloan: size / rate	≤\$50k (avg ~\$13–16k) / 8–13%	SBA (primary)
Peninsula industrial rent / smallest units	\$1.98–2.48/SF/mo NNN / ~800 SF	Broker research + aggregators
PG&E B-1 / residential TOU blended (with PCE)	\$0.41–0.42 / \$0.38–0.40 per kWh	Tariff sheets / CCA comparisons
Redwood City sales tax / minimum wage	9.875% / \$18.65 per hr	CDTFA-concordant / city
CA: LLC tax / \$179 cap / bonus / QBI	\$800 / \$25k / none / none	FTB (primary)
Federal: bonus / \$179 / de minimis (2026)	100% / \$2.56M / \$2,500-per-item	OBBBA, Rev. Proc. 2025-32
Mfg partial sales-tax exemption	–3.9375 pts through 6/30/2030	CDTFA RTC 6377.1 (primary)
Fleet-SaaS leader ARR / pricing band	~\$350k / \$4–40 per printer-mo	Company profile / vendor pricing

9.2 Method

Five specialist analyses (cost accounting, capital budgeting, utilization/throughput, tax, ownership risk) were drafted independently, then attacked by an adversarial credit-officer review that recomputed the arithmetic and re-verified market claims. A deep-research pass decomposed the Redwood City question into five search angles, fetched 23 sources, extracted 40 falsifiable claims, and subjected the top 25 to three-vote adversarial verification (2/3 refutations kill a claim). Three

targeted fact-check sweeps then closed the gaps verification exposed. Where primary sites blocked automated retrieval (FTB, CDTFA, redwoodcity.org, pge.com), figures were confirmed through multiple concordant secondary sources and are flagged accordingly in the underlying plan document.

9.3 Claims that failed verification — excluded from all math above

- "\$10,000/month from 4–6 well-utilized printers" (vendor-derived) — refuted; top-decile, not base case.
- "B2B prototyping standardly bills \$100+/design-hour" — refuted as a general anchor.
- "Queue software saves 10–20 hrs/week and doubles capacity" (vendor survey) — refuted as independent fact.
- "Bay Area turnaround bar is under one day, which entrants must beat" — refuted; same-day is a differentiator, not table stakes.
- "Fleet standardization is a precondition for automation ROI" — failed verification as stated.

Companion document: *3d-printing-business-plan.md* (same repository branch) carries the full narrative plan, source URLs, and the pre-launch compliance checklist. Figures in this report were generated from the formulas and inputs shown; the figure-generation script is reproducible on request.